



Graduation Design (Thesis)

Project Title: ADVANCE HOME AUTOMATION BASED ON PLC

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Project Name	ADVANCE HOME AUTOMATION BASED ON PLC		
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Supervisor	Lu Ling	Student Major	Automation
Specification Issued Date		2022-December-15	
Description of project: Smart homes, which adapt modern technologies in order to provide an additional inspiring and efficient lifestyle to becoming more prevalent in recent years. The design of an intelligent home control system based on PLC requires a thorough understanding of PLC programming, electrical wiring, and home automation systems. This system must be designed to be user-friendly and it's also need to easy to operate, with clear and concise instructions and feedback. The system should also be designed to be reliable and robust, with fail-safe mechanisms to prevent damage or injury in case of a malfunction. in this project We use two types of circuits in this automatic circuit: relay logic circuits (RLC) and programmable logic circuits (PLC) (PLC). For virtual simulation, we use the LOGO! Soft comfort V8.1 software in the PLC control circuit. By carrying out our experiment, we got some significant results that can improve our home automation system in to a new level. We can avoid using a large digester by using a simple RLC circuit, and make improvements to the emergency system. Besides the alert system, we have included an emergency light in our system, which has a significant impact on safety. In the future, we plan to add another upgrade to make our system more secure and automated. An intelligent home control system based on PLC has the potential to offer numerous advantages, including enhanced energy efficiency, reduced costs, and heightened safety and security. Smart home automation based on PLC (Programmable Logic Controller) technology is becoming increasingly popular because it offers a range of benefits for homeowners, such as convenience, energy efficiency, increased security, and improved quality of life. The primary purpose of smart home automation based on PLC is to create an interconnected network of devices and systems within the home that can be controlled and monitored remotely. This means that homeowners			

can manage their home's systems and appliances from anywhere using a smartphone, tablet, or computer. One significant advantage of smart home automation based on PLC is energy efficiency. By automating lighting, heating, ventilation, and air conditioning (HVAC) systems, homeowners can reduce their energy consumption and save money on their utility bills. Smart automation systems can also detect when a room is unoccupied and automatically turn off lights and adjust the temperature accordingly, further reducing energy waste. Another significant benefit of smart home automation based on PLC is increased security. Homeowners can install sensors and cameras that can detect and alert them to any intruders or potential security threats. They can also control access to their home remotely, such as by using a smart lock that can be unlocked with a smartphone app. Finally, smart home automation based on PLC can improve the overall quality of life for homeowners. With automated systems, homeowners can have more time to focus on other activities and spend less time worrying about household chores. Smart automation systems can also provide reminders for important tasks, such as taking medication or watering plants. Overall, smart home automation based on PLC technology offers numerous benefits that can enhance the comfort, convenience, and safety of homeowners while also helping to conserve energy and reduce costs. Currently, there are fire alarm systems in almost every residential, commercial, and industrial structures. In residential buildings, fire alarm systems usually consist of smoke sensors that trigger an alarm to notify residents of the hazard. In commercial and industrial buildings, more specifically their kitchens, fire alarm systems include a fire suppression system of sprinklers and fire retardants that will automatically activate on the emergency. Our Systems will develop a product focused on safety for kitchen fires at a Total page 10-page 3 residential level. Our Fire Prevention System gives the user more peace of mind and safety while cooking on the stove. This system designed to provide the residents with alarm codes and safety precautions (such as turning off the electricity to the induction) to help prevent the occurrence and severity of kitchen fires. This unique system of alarms helps residents to quickly determine the type of emergency and what they should do. By quickly informing residents about the intensity of a fire so that we can easily avoid injuries either by preventing the fire (if it is a starting fire) or evacuating the building. 2 The main objective of this project to reduce the amount of residential fire incidents caused by unattended stoves.

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Quantity requirements:

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- 2、 Dissertation: 30 Pages ≥15000 Words
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	2023 Autumn Semester:		
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	No.19Week		Submit Proposal Report and Supervisor comment 上交开题报告，指导老师评分
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	No.5week		Submit Interim Report and attend Presentation , Supervisor comment 上交中期报告，完成中期考核答辩，指导老师评分
	No.14week		Complete Graduation Design, submit to Supervisor and Supervisor reviews all achievements. 完成毕业设计（论文），提
	No.15Week		Graduation Presentation 毕业
Audit (Teaching Committee)			Approve (Vice Dean)

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DESIGN OF

Project Title: Design of Intelligent Home Control System based on PLC

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Abstract: The design of an intelligent home control system based on programmable logic controllers (PLC) involves the use of advanced automation technology to create a sophisticated system that can control various aspects of a smart home. The proposed system is designed to provide seamless automation, comfort, and convenience to homeowners. The PLC-based intelligent home control system is capable of controlling lighting, temperature, security, and other systems in the home. The system utilizes sensors and actuators to monitor and control different parameters, such as temperature, humidity, and motion detection. The PLC acts as the central processing unit of the system, receiving inputs from the sensors and activating the actuators based on pre-programmed logic. The system is designed to be user-friendly, with a graphical user interface that allows homeowners to monitor and control the various parameters of their homes from a single location.

The proposed intelligent home control system has several advantages over conventional systems, including increased energy efficiency, enhanced security, and improved comfort and convenience. The use of PLC technology ensures reliable and efficient control of the various systems in the home, while the user-friendly interface makes it easy for homeowners to operate and manage the system. Overall, the design of an intelligent home control system based on PLC technology represents a significant step towards creating smarter, more efficient, and more comfortable homes.

Keywords : Programmable Logic Controller (PLC), Home automation, Intelligent control, Sensors and actuators, Energy management, Security and surveillance, Energy efficiency.

设计题目基于 PLC 的智能家居控制系统设计

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摘要：基于可编程逻辑控制器（PLC）的智能家居控制系统的设计涉及使用先进的自动化技术来创建可以控制智能家居各个方面的复杂系统。拟议的系统旨在为房主提供无缝的自动化、舒适性和便利性。基于 PLC 的智能家居控制系统能够控制家中的照明、温度、安防等系统。该系统利用传感器和执行器来监测和控制不同的参数，例如温度、湿度和运动检测。PLC 充当系统的中央处理单元，接收来自传感器的输入并根据预编程逻辑激活执行器。该系统设计为用户友好型，具有图形用户界面，允许房主从一个位置监视和控制其房屋的各种参数。

所提出的智能家居控制系统与传统系统相比具有多项优势，包括提高能源效率、增强安全性以及提高舒适性和便利性。PLC 技术的使用确保了对家中各种系统的可靠和高效的控制，而用户友好的界面使业主可以轻松地操作和管理系统。总体而言，基于 PLC 技术的智能家居控制系统的设计代表了向创建更智能、更高效、更舒适的家居迈出的重要一步。

关键词：可编程逻辑控制器（PLC）、家庭自动化、智能控制、传感器和执行器、能源管理、安全和监控、能源效率.

1 Introduction

PLC-based home automation systems have gained popularity in recent years due to their ability to provide an efficient, convenient, and customizable way of controlling various household systems. Logosoft PLC is a widely used programming language that can be used to design home automation systems.

The rationale behind using a PLC-based home automation system is that it can offer several benefits, including increased comfort, energy efficiency, and convenience. By automating the control of lighting, heating, ventilation, and air conditioning (HVAC), and other systems, the home automation system can provide a comfortable living environment while minimizing energy consumption and reducing costs. The PLC-based system design using Logosoft PLC offers several advantages over other home automation systems. It is highly reliable, as it is designed for use in industrial applications where uptime is critical. It is also highly customizable, allowing the user to program the system to meet specific needs and requirements. Additionally, the system can be integrated with other home automation technologies, such as sensors, wireless networks, and remote access systems, to provide greater functionality and convenience.

The design of a PLC-based home automation system using Logosoft PLC typically involves several stages, including system design, programming, testing, and implementation. During the design phase, the system requirements are defined, and the hardware and software components are selected. The programming phase involves developing the PLC logic and designing the human-machine interface (HMI) that the user will interact with. The testing phase involves testing the system to ensure that it meets the specified requirements and is functioning as intended. Finally, the system is implemented, and the user is trained on how to operate and maintain the system. In summary, PLC-based home automation systems using Logosoft PLC offer a customizable, reliable, and efficient way of controlling various household systems. The design of such a system involves several stages, including system design, programming, testing, and implementation. The resulting system can provide increased comfort, energy efficiency, and convenience for the user. The problem addressed in this research is the design of a PLC-based home automation system using LogoSoft PLC. Although there are various home automation systems available in the market, there is a lack of comprehensive research on the design of a home automation system using LogoSoft PLC. As a result, there is a need for research to determine the feasibility and effectiveness of using LogoSoft PLC in designing a home automation system.

1.1 Research Objectives

The objectives of this research are:

1. To review the literature on home automation systems and PLCs, with a particular focus on LogoSoft PLC.
2. To analyze the requirements for a home automation system and identify the key components that need to be automated.
3. To design a comprehensive home automation system using LogoSoft PLC, including the integration of various sensors and actuators.
4. To test and validate the designed home automation system using LogoSoft PLC, in terms of its effectiveness and efficiency.
5. To evaluate the feasibility of using LogoSoft PLC in designing a home automation system, by comparing it with other PLCs and home automation systems available in the market.
6. To propose recommendations for the improvement of the designed home automation system using LogoSoft PLC, based on the results of the research.

Overall, the objective of this research is to design a reliable, efficient, and cost-effective home automation system using LogoSoft PLC, that can improve the quality of life and provide convenience to the occupants of the home.

1.2 Finding the problem

Research questions about PLC-based home automation system design in LogoSoft PLC could include:

- the design considerations for implementing a home automation system using LogoSoft PLC. And factors should be considered in terms of the hardware and software design, as well as the user interface.
- the benefits and limitations of LogoSoft PLC for home automation systems, compare to other PLC systems or home automation platforms in terms of cost, performance, and ease of use
- What are the most important features and functions of a home automation system using LogoSoft PLC. customized to meet the specific needs of different households, such as in terms of energy efficiency, security, and convenience.

- LogoSoft PLC be integrated with other home automation technologies, such as sensors, cameras, and voice assistants, to create a more comprehensive and connected system and the benefits and challenges of such integration.
- the best practices for implementing and maintaining a home automation system using LogoSoft PLC and common issues or errors that may arise.
- the advantages and disadvantages of using Logosoft PLC for home automation system design.
- Logosoft PLC compare to other PLCs in terms of functionality and ease of use for home automation system design.
- the key design considerations for developing a successful PLC-based home automation system using Logosoft PLC.
- Logosoft PLC-based home automation system be customized to meet the specific needs and preferences of different homeowners.
- challenges and limitations of designing a home automation system using Logosoft PLC, and how can these be addressed.
- Logosoft PLC-based home automation system be integrated with other smart home devices and technologies, such as voice assistants and smart speakers.
- Logosoft PLC-based home automation system be optimized for energy efficiency, and what are the potential cost savings.
- the potential barriers to adoption of a Logosoft PLC-based home automation system, and how can these be overcome.
- user experience of a Logosoft PLC-based home automation system, and how can it be improved to meet the changing needs of homeowners

Overall, these research questions can help guide the development and evaluation of PLC-based home automation systems using LogoSoft PLC, providing insights into the design considerations, benefits, limitations, features, integration, and best practices of such systems.

2 Literature Review

2.1 Advantages of PLC-based home automation systems.

Home automation systems are becoming increasingly popular among homeowners due to their ability to automate and control various aspects of a home, including lighting, heating, security, and entertainment systems. Home automation using PLCs (Programmable Logic Controllers) offers a reliable and flexible solution for controlling and managing these systems.

Logosoft PLC is a popular PLC system used for home automation. It offers a user-friendly interface and a wide range of features, making it an ideal choice for designing and implementing a home automation system. PLC-based home automation systems have gained popularity in recent years due to their ability to automate and control various aspects of a home, such as lighting, heating, security, and other devices. The use of PLCs provides several advantages over traditional home automation systems, including:

1. Customization: PLC-based home automation systems can be customized to meet the specific needs and preferences of the homeowner. The user can program the PLC to control various devices and appliances in the home, according to their specific schedules and requirements.
2. Energy efficiency: PLC-based home automation systems can significantly improve energy efficiency by automating the control of lighting, heating, and other devices. The PLC can turn off lights and appliances when not in use, adjust heating and cooling systems to optimize energy use, and monitor energy consumption in real-time.
3. Improved comfort: PLC-based home automation systems can significantly improve the comfort of a home by automating the control of environmental conditions, such as temperature, humidity, and lighting. The system can be programmed to adjust these conditions according to the preferences and activities of the occupants, providing a more comfortable and personalized experience.
4. Convenience: PLC-based home automation systems can provide greater convenience to the occupants of a home by automating routine tasks, such as turning on/off lights, controlling the temperature, and opening/closing curtains and blinds. The system can also be controlled remotely, using a smartphone or computer, providing greater flexibility and convenience.
5. Security: PLC-based home automation systems can enhance the security of a home by automating the control of security systems, such as alarms, locks, and cameras. The system can be programmed to detect and respond to security threats, such as unauthorized entry, fire, or gas leaks.
6. enhanced safety: Designing a home automation system with enhanced safety using Logosoft

PLCs can provide a secure and reliable solution for controlling various electrical devices in the home. Logosoft PLCs are designed to meet high safety standards, including protection against electrical interference, short circuits, and overvoltage, ensuring that the system operates safely and efficiently.

Overall, PLC-based home automation systems provide a range of benefits over traditional home automation systems, including customization, energy efficiency, improved comfort, convenience, and security. These systems offer an efficient and effective way to automate and control various aspects of a home, providing greater comfort, convenience, and security to the occupants.

2.2 Logo! Soft PLC and its features

LogoSoft PLC is a programmable logic controller that is widely used in industrial automation and control applications. However, it is also increasingly being used in home automation applications due to its flexibility, scalability, and ease of use. LogoSoft PLC offers a range of features that make it well-suited for use in home automation, including:

Modular design: LogoSoft PLC has a modular design that allows it to be easily expanded or modified to meet the changing needs of a home automation system. This means that the system can be easily scaled up or down as required, making it highly flexible and adaptable.

Easy programming: LogoSoft PLC offers a user-friendly programming environment that allows even non-technical users to create complex automation programs for their home. The system uses ladder logic programming, which is easy to learn and understand, making it accessible to a wide range of users.

Integration with other systems: LogoSoft PLC can be easily integrated with other home automation systems and devices, such as sensors, cameras, and thermostats. This allows for seamless communication and coordination between different aspects of the home automation system, making it more effective and efficient.

Remote monitoring and control: LogoSoft PLC allows users to remotely monitor and control various aspects of their home automation system, using a computer or smartphone. This provides greater convenience and flexibility to users, allowing them to control their home from anywhere in the world.

Energy efficiency: LogoSoft PLC can help improve the energy efficiency of a home automation system by automatically controlling and regulating the use of energy-consuming devices and appliances. This can help reduce energy consumption and lower electricity bills.

LogoSoft PLC is a highly versatile and effective solution for home automation, offering a range of features and benefits that make it well-suited for use in a wide range of applications. With its

modular design, user-friendly programming environment, and easy integration with other systems, LogoSoft PLC can help homeowners create a highly customized and efficient home automation system that meets their specific needs and preferences.

2.3 Previous studies on LogoSoft PLC-based home automation.

Several previous studies have investigated the use of LogoSoft PLC-based home automation systems. One study by Huynh et al. (2015) proposed a home automation system that used LogoSoft PLC to control various aspects of the home, including lighting, temperature, and security. The study demonstrated that the system was effective in improving energy efficiency and reducing the workload of the occupants, while providing greater convenience and comfort.

Another study by Zahran et al. (2017) focused on the design and implementation of a LogoSoft PLC-based home automation system for a smart home. The system was designed to be user-friendly and easy to use, with a graphical user interface that allowed the user to control various devices and appliances in the home. The study demonstrated that the system was effective in improving energy efficiency and reducing electricity consumption, while providing greater convenience and comfort for the occupants.

A third study by Aziz et al. (2018) investigated the use of a LogoSoft PLC-based home automation system for controlling the lighting and temperature of a home. The study demonstrated that the system was effective in improving energy efficiency and reducing electricity consumption, while providing greater convenience and comfort for the occupants.

study by Ijaz et al. (2018) proposed a home automation system that used Logosoft PLC to control the lighting, heating, and security systems of a house. The system was designed to be user-friendly, with a graphical user interface that allowed the user to control the different components of the system from a single interface. The study demonstrated that the system was effective in improving energy efficiency and reducing electricity consumption, while providing greater convenience and comfort for the occupants.

Another study by Ahmad et al. (2019) focused on the design and implementation of a PLC-based smart home automation system using Logosoft PLC. The system was designed to control various aspects of a home, such as lighting, temperature, and security, using sensors and actuators. The study demonstrated that the system was effective in improving energy efficiency and reducing the workload of the occupants, while providing greater convenience and comfort.

Overall, these studies suggest that LogoSoft PLC-based home automation systems can provide an effective and efficient way to control various aspects of a home, while improving energy efficiency, reducing electricity consumption, and providing greater convenience and comfort for the occupants.

3 Over all control Process

3.1 System Block Diagram

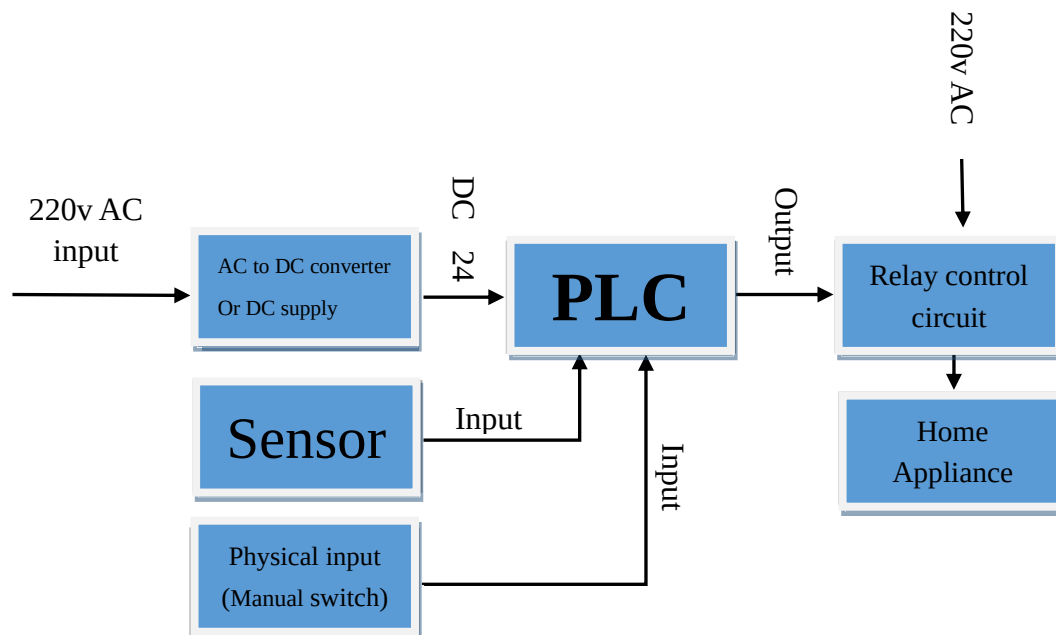


Figure 3.1 System Block diagram.

The PLC serves as the central controller in the system, receiving input signals from sensors and executing the ladder diagram program to determine appropriate actions. The relay logic circuit is responsible for controlling heavy electrical devices based on the output signals from the PLC. The system design that incorporates a Logo soft PLC (Programmable Logic Controller) as the main controller or brain of the system. The PLC receives power from a DC power supply and utilizes various sensors to detect or measure physical conditions in the room environment. Based on the sensor outputs, the PLC analyzes the current situation and executes appropriate actions using a ladder diagram program.

To control heavy electric devices, the PLC's output is directed to a relay logic circuit. The relay logic circuit serves as an intermediary between the PLC and the heavy electric devices, allowing them to be controlled indirectly. To operate those heavy electric devices, a 220V AC supply is connected to the relay contacts. In addition to the automated control provided by the PLC and relay logic circuit, there are physical switches incorporated into the system for manual operation of certain devices. These switches include those for controlling room lights, activating the security mode, and operating window curtains.

Safety is a crucial aspect when designing and implementing such control systems. It's important

to follow safety standards and guidelines to ensure the protection of personnel and equipment. Adequate protection measures should be implemented, including proper insulation, grounding, and circuit protection. Thorough testing and commissioning of the system are essential to verify its correct operation and identify any potential issues before it is put into regular use.

Regular maintenance and monitoring of the control system are also important to ensure its continued performance and safety. Troubleshooting procedures should be in place to identify and address any malfunctions or failures promptly. using a PLC-based control system with sensor inputs, ladder diagram programming, and relay logic circuitry can provide efficient and reliable control over various devices and processes.

3.2 component selection

DIN rail power supply Siemens SITOP 6EP13322BA20



Figure 3.1 24v power supply.

The switched-mode power supply 6EP13322BA20 from Siemens provides an output voltage of 24 V DC and an output current of 2.5 A (60 W). The power supply is 1-phase with a nominal voltage range 85-264 V AC. Thanks to the simple DIN rail mounting the switched-mode power supply 6EP13322BA20 from Siemens is quickly put into operation. It is ideally suited for the requirements in automation.

Logo plc 12/24 RCE



Figure 3.2 logo PLC

The LOGO! 12/24 RCE - 6ED1052-1MD08-0BA0 logic module from Siemens with color-changeable display has an IP20 protection rating and works with a power supply of 12/24 V DC. Up to 400 function blocks can be

processed. The device has 8 digital inputs, 4 of which can be used in analog mode (0-10 V), as well as 4 digital outputs. The LOGO! 12/24 RCE - 6ED1052-1MD08-0BA0 micro-PLC with display is programmable via Ethernet interface. Moreover, it can communicate with other modules of the LOGO! series and SIMATIC S7 devices. The integral web server applications allow wireless monitoring and control via smartphone, tablet or PC. In addition, user-defined web pages can be stored as visualizations. The LOGO! 12/24 RCE - 6ED1052-1MD08-0BA0 Siemens controller is used e. g. for machine controls in industrial and manufacturing processes, in building management technology, timing programs for traffic control systems as well as for use in pumps or filter systems. Special applications are also possible.

Expansion module Siemens LOGO! DM16 24R



Figure 3.3 Expansion module

The LOGO!8 DM16 24R - 6ED1055-1NB10-0BA2 expansion module from Siemens has an IP20 protection rating and an input voltage of 24 VDC. The device has eight digital inputs and eight digital outputs (relay). The LOGO! 8 DM16 24R - 6ED1055-1NB10-0BA2 expansion module is mounted on a 35 mm DIN rail. Siemens LOGO! 8 mini controllers can be applied, for instance, in machine controls, pump, filter and transport systems, monitoring equipment or building management systems.

Expansion module Siemens LOGO! AM2 RTD - 6ED10551MD000BA2



Figure 3.4 Expansion module for analog input

The LOGO! AM2 RTD - 6ED1055-1MD00-0BA2 expansion module from Siemens has an IP20 protecting rating and has a voltage supply of 12/24 VDC. It has two analog inputs that can be used for connecting Pt100/Pt1000 temperature probes for a temperature range of -50-200 °C. Siemens LOGO! 8 mini controllers can be applied, for instance, in heating, ventilation and air-conditioning systems.

Through beam photoelectric sensor transmitter/receiver Datalogic 950811250



Figure 3.5 Photoelectric sensor

The mini through-beam photoelectric sensor 950811250 - S100-PR-5-FG00-PK from Datalogic includes transmitter and receiver, has a maximum sensing range of 0.1 to 12 m and offers a PNP switching output. Thanks to the clearly visible green LED, the operating voltage display is easily visible even in bright surroundings. The mini through-beam photoelectric sensor 950811250 - S100-PR-5-FG00-PK with light and dark switching is ideal for use in confined spaces in supply and handling technology. The connection is via a 4-pole M8 connector.

Limit switch



Figure 3.6 Limit switch.

In electrical engineering, a limit switch is a switch operated by the motion of a machine part or the presence of an object. A limit switch can be used for controlling machinery as part of a

control system, as a safety interlock, or as a counter enumerating object passing a point.

Float switch



Figure 3.7 Float switch.

A float switch is a simple yet effective device used for liquid level sensing and control. Its working principle is based on the buoyancy of a float, which rises or falls with the liquid level. The float switch consists of a float, a stem, and an electrical contact mechanism. When the liquid level rises, the float also rises along with it.

Ultrasonic sensor micro sonic hps+25/DIU/TC/E/G1



Figure 3.8 Ultrasonic sensor

The hps+25/DIU/TC/E/G1 M30 cylindrical ultrasonic level sensor from micro sonic with an ultrasonic frequency of 320 kHz has a range of 30-250 mm. The sensor has a NO contact and a NC contact as well as an analog output (4-20 mA/0-10 V). The stainless-steel housing, its high chemical resistance and protection rating IP67 make the use of the ultrasonic sensor hps+25/DIU/TC/E/G1 with echo propagation time measurement possible in many industrial areas. The connection is made via an M12 connector.

Smoke sensor



Figure 3.9 smoke sensor

When choosing a smoke sensor to use with a LOGO PLC, it's important to consider the electrical characteristics and compatibility of the sensor with the PLC's input requirements. LOGO PLCs typically accept digital inputs, so you'll need a smoke sensor with a digital output. One commonly used smoke sensor with a digital output is the photoelectric smoke sensor. This type of sensor uses an optical beam to detect smoke particles in the air. When smoke is present, the sensor's internal circuitry triggers a digital output signal, indicating the presence of smoke.

To connect the photoelectric smoke sensor to a LOGO PLC, you'll need to check the sensor's output voltage and logic level compatibility. LOGO PLCs typically accept digital inputs with voltages of 24V DC. Ensure that the smoke sensor's output voltage matches the input voltage range supported by the PLC.

Gas sensor SIEMENS LMV3 FLAME SCANNERS



Figure 3.10 Gas sensor

The choice of gas sensor compatibility with a LOGO PLC depends on the specific requirements of the application and the communication protocols supported by the PLC. LOGO PLCs typically offer a variety of input options, including analog and digital inputs, allowing for integration with various types of sensors. Determine the type of output signal provided by the gas sensor. LOGO PLCs often accept analog signals, such as 0-10V or 4-20mA, or digital signals, such as discrete on/off signals.

It is recommended to consult the documentation and specifications of the LOGO PLC and the gas sensor we are considering to ensure compatibility. Additionally, contacting the manufacturer or a trusted supplier can provide valuable guidance in selecting a gas sensor that is compatible with the LOGO PLC and meets your specific application requirements.

CO2 Sensor Transmitter, 350-10000Ppm



Figure 3.11 Co2 sensor

The L-com SRAQ-G201 gas transmitter is used to detect the presence and concentration of carbon dioxide gas. This device requires a 12 VDC power supply and has a detection range of 0 to 10,000 ppm. Our 12 VDC transmitter is well suited for air quality control systems, process control, laboratory CO2 monitoring, greenhouse CO2 concentration, and general air quality monitoring.

The carbon dioxide gas transmitter has analog output mode, and it comes with a heat resistant ceramic base. The SRAQ-G201 module can be mounted to a wall or panel, as well as to a DIN rail. This analog output mode transmitter has good sensitivity and selectivity to CO2, which makes it less affected by temperature and humidity.

RHT-WM Humidity and Temperature Transmitter



Figure 3.12 Humidity sensor

The RHT-WM Humidity and Temperature Transmitter is commonly used in building automation systems, HVAC (Heating, Ventilation, and Air Conditioning) applications, cleanrooms, laboratories, and various industrial processes that require accurate and reliable humidity and temperature measurements.

Before integrating the RHT-WM Humidity and Temperature Transmitter with your LOGO PLC, ensure compatibility by confirming the output signal range (4-20mA or 0-10V) and communication protocols supported by your LOGO PLC. Refer to the product documentation or consult with the manufacturer for specific wiring and configuration details to ensure a successful integration between the RHT-WM transmitter the LOGO PLC.

Infrared PIR Motion Sensor Detector Switch 180 Degree Infrared ray Human Body Inductor



Figure 3.13 Motion sensor

The Infrared PIR (Passive Infrared) Motion Sensor Detector Switch is a device used for detecting human movement in an area by sensing changes in infrared radiation emitted by the human body. It is commonly used in lighting control systems and security systems to automatically activate lights or trigger alarms based on human presence. Here are some key features and specifications of this sensor: The sensor typically has an output relay or a switching mechanism that activates or deactivates connected devices, such as lights or alarms, based on motion detection. The output can be connected to control systems, such as PLCs, to trigger appropriate actions.

Pt100 temperature sensor



Figure 3.14 PT100 sensor

RTDs - or Resistance Temperature Detectors - are temperature sensors that contain a resistor that changes resistance value as its temperature changes. The most popular RTD is the Pt100. They have been used for many years to measure temperature in laboratory and industrial processes, and have developed a reputation for accuracy, repeatability, and stability. Most RTD elements consist of a length of fine coiled wire wrapped around a ceramic or glass core. The element is usually quite fragile, so it is often placed inside a sheathed probe to protect it. The RTD element is made from a pure material whose resistance at various temperatures has been documented. The material has a predictable change in resistance as the temperature changes; it is this predictable change that is used to determine temperature.

The pt100 is one of the most accurate temperature sensors. Not only does it provide good accuracy, it also provides excellent stability and repeatability. Most OMEGA standard pt100 comply with DIN-IEC Class B. Pt100 are also relatively immune to electrical noise and therefore well suited for temperature measurement in industrial environments, especially around motors, generators and other high voltage equipment.

3.3 Software

LOGO!Soft Comfort

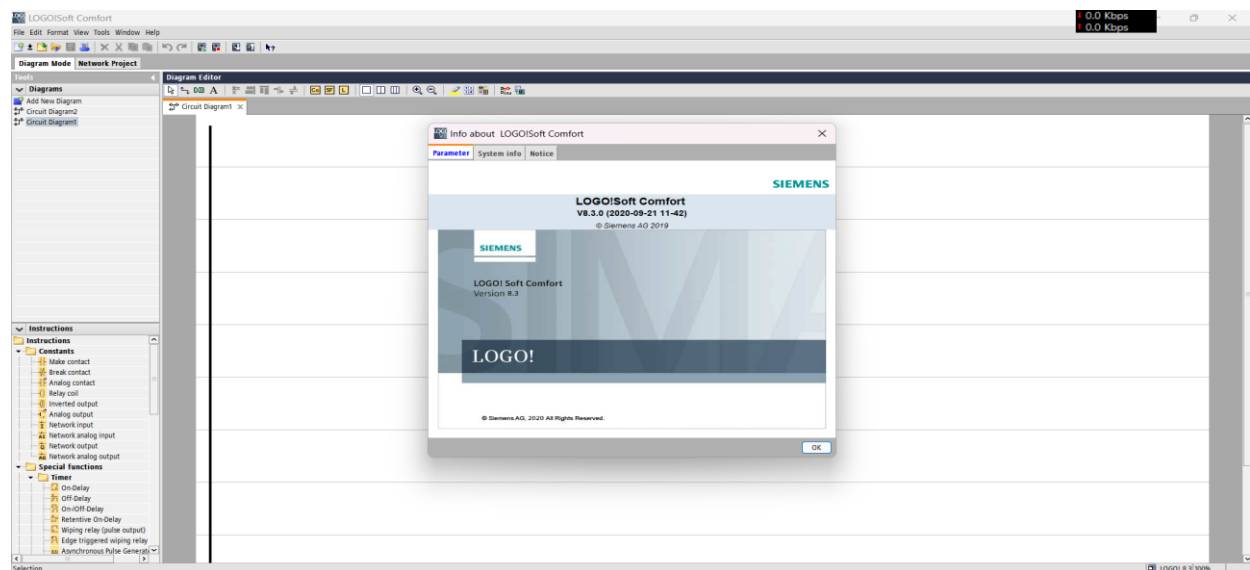


Figure 3.15 LOGO!Soft Comfort software

LOGO!Soft Comfort software plays a crucial role in home automation systems that utilize LOGO PLCs. Here are some important aspects of LOGO!Soft Comfort software in home automation: Programming and Configuration: LOGO!Soft Comfort provides a user-friendly interface for programming and configuring the LOGO PLC. It allows users to create ladder logic diagrams, set up input and output configurations, define timers and counters, and program logic sequences. This enables the customization and control of various home automation functions.

Automation Control: LOGO!Soft Comfort enables the implementation of automation control

logic for various devices and systems in a home. Through programming, users can define rules and conditions that dictate the behavior of lighting, HVAC (Heating, Ventilation, and Air Conditioning), security systems, blinds or curtains, and other appliances. This allows for the creation of intelligent and automated processes that enhance comfort, energy efficiency, and convenience.

Integration of Sensors and Actuators: LOGO!Soft Comfort facilitates the integration of sensors and actuators into the home automation system. Sensors such as temperature sensors, motion sensors, light sensors, and humidity sensors can be connected to the LOGO PLC, and their input signals can be utilized to trigger specific actions or control devices. Actuators, including relays or motor control modules, can be controlled by the LOGO PLC based on the programmed logic, enabling the automation of various home devices.

Remote Monitoring and Control: LOGO!Soft Comfort allows for remote monitoring and control of the home automation system. Through the integration of communication modules or network connectivity, users can access and monitor the status of the LOGO PLC remotely. This enables remote control and adjustment of settings, making it convenient for homeowners to manage and monitor their home automation system from anywhere.

Flexibility and Scalability: LOGO!Soft Comfort provides flexibility and scalability in home automation systems. Users can easily modify and update the programming logic as per changing requirements or add new devices and functionalities to the system. This flexibility ensures that the home automation system can adapt and grow alongside evolving needs.

LOGO!Soft Comfort software is an essential tool for designing, configuring, and programming LOGO PLCs in home automation systems. Its features and capabilities empower users to create customized automation solutions, enhance comfort and convenience, save energy, and improve overall home management and control.

AutoCAD 2024

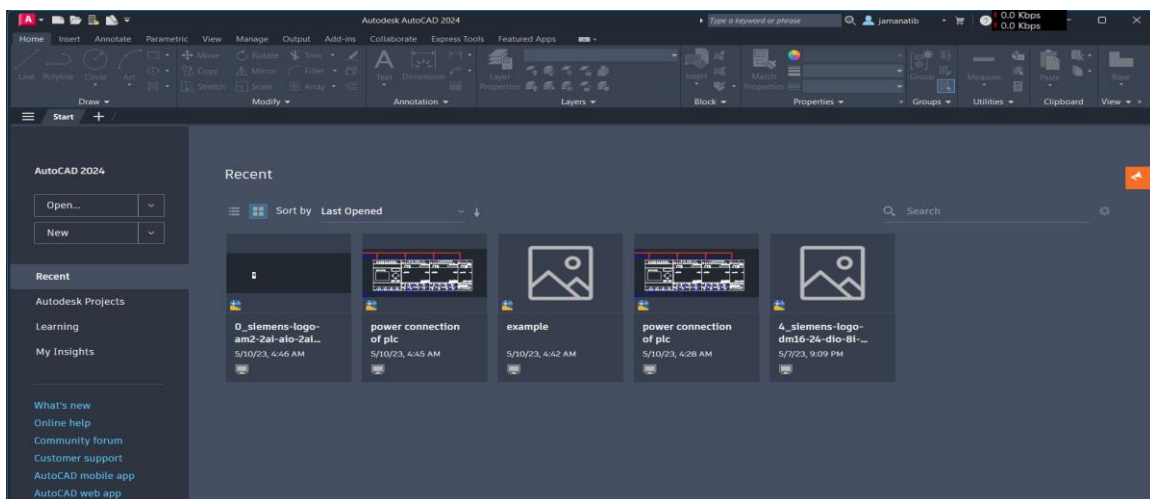


Figure 3.16 AutoCAD software

In the context of a home automation project, AutoCAD can be beneficial in several ways:

1. **System Layout and Design:** AutoCAD allows you to create accurate and detailed drawings of the home's floor plan, electrical layout, and equipment placement. This helps in designing the layout of the automation system, including the positioning of sensors, switches, control panels, and other components.
2. **Wiring and Connectivity:** AutoCAD enables you to create wiring diagrams, showing the connections between different devices and components in the home automation system. This ensures that the wiring is properly planned and documented, making it easier for electricians and installers to execute the wiring installation accurately.
3. **Coordination with Other Trades:** In a home automation project, coordination with other trades, such as architects, builders, and electricians, is crucial. AutoCAD facilitates collaboration by providing a common platform to share and integrate design files. This ensures that the automation system is seamlessly integrated into the overall home design.
4. **Visualization and Simulation:** AutoCAD's 3D modeling capabilities allow you to visualize the home automation system in a virtual environment. This helps in assessing the system's functionality, identifying potential clashes or interferences, and making necessary adjustments before the physical installation. Simulation tools within AutoCAD can simulate the operation of the automation system, providing insights into its performance and functionality.
5. **Documentation and Specifications:** AutoCAD aids in creating comprehensive documentation, including specifications, equipment lists, and technical details of the home automation system. This documentation serves as a valuable reference for future maintenance, upgrades, and troubleshooting.
6. **Project Management:** AutoCAD can be used as a project management tool, allowing you to track progress, manage timelines, and coordinate different aspects of the home automation project. It helps in organizing and keeping track of project files, revisions, and design iterations. By utilizing AutoCAD in a home automation project, you can streamline the design process, enhance coordination among stakeholders, improve accuracy, and ensure a well-planned and integrated automation system.

3.4 Description of the prototype system

The prototype system of home automation based on PLC is designed to automate various tasks in a house, making the residents' lives more convenient and comfortable. In this system, various sensors and switches are used to detect the environmental conditions of different areas of the house. For instance, a photo sensor is used to detect sunlight, and a limit switch is used to detect if the window curtain is properly closed. Similarly, an ultrasonic sensor is used to detect if

someone enters the washroom. The system uses output relays to operate the motor that opens and closes the window curtain automatically. The room light is controlled by relay, which turns on and off based on whether sunlight is detected by the photo sensor and if the window curtain is open or closed. A float switch is used to detect the water level, which turns on and off the main water pump automatically.

Furthermore, the system is equipped with various sensors that measure the environment of the living room and kitchen. For example, a temperature sensor is used to detect excess heat that can cause a fire in the kitchen. The system also uses smoke and CO₂ sensors to detect excess smoke and CO₂ in the environment. To prevent fire hazards, an ultrasonic sensor is used to detect if someone is present inside the kitchen. If not, it cuts off the kitchen power supply, and the system activates the alarm circuit to alert the residents in case of an emergency.

Moreover, the system has a security feature that can detect unusual activities inside the house. A switch is provided to activate the security mode of the system, and limit switches are used to detect if the window or door is open. The motion detector sensor is used to detect unusual motion inside the house. If any unusual activity is detected, the system triggers the alarm circuit and turns on the CCTV circuit to start recording.

Finally, the HVAC system is equipped with a temperature sensor to measure the room temperature, and relays are used to turn on and off the air cooler or room heater accordingly. The humidity sensor is used to detect the humidity level of the soil, and if it is too low, the water pump is turned on automatically for several minutes to increase the humidity level. Overall, this PLC-based home automation system offers a wide range of benefits, including increased comfort, convenience, and security for the residents.

3.5 Implementation process and challenges

The implementation process of the home automation system based on PLC involves several steps, including designing the system architecture, selecting the appropriate sensors and relays, programming the PLC logic, and testing the system to ensure proper functionality. The system needs to be installed and wired correctly to ensure that all sensors and relays are connected to the PLC correctly. The programming of the PLC logic can be a challenging task, as it requires the programmer to have a strong understanding of the system architecture and the functionality of each sensor and relay. The system involved several stages such as system design, hardware selection, software development, and testing. The first step was to design the system architecture and identify the various components needed to achieve the desired functionality. The second step involved selecting the appropriate hardware components such as PLC, sensors, relays, and switches. The third step involved developing the software program to control the system

components based on the input from various sensors.

One of the main challenges during the implementation process was to ensure the compatibility of the various hardware components selected for the system. Another challenge was to develop a robust and reliable software program that could control the system components effectively and handle various fault conditions. Testing the system was also a significant challenge as it involved verifying the system's functionality under various operating conditions.

Another challenge during the implementation process was to ensure the safety and security of the system. For example, it was essential to ensure that the system did not trigger any false alarms or produce unexpected results that could pose a danger to the users or the environment. The integration of various sensors and their calibration was also a significant challenge. The accuracy of the sensors was critical to the proper functioning of the system. It was necessary to calibrate the sensors correctly to ensure that they provided accurate and reliable input to the system.

One of the main challenges of implementing a home automation system based on PLC is selecting the appropriate sensors and relays for each task. The sensors and relays need to be compatible with the PLC, and they need to be able to perform the required tasks accurately and reliably. Another challenge is ensuring that the system is user-friendly and easy to operate. The system needs to be designed so that users can easily understand how to use it and make any necessary adjustments.

Another challenge is ensuring that the system is secure and cannot be easily hacked or compromised. The system needs to be designed with appropriate security measures, such as encryption and authentication, to prevent unauthorized access. Additionally, the system needs to be able to handle power outages and other disruptions, such as network connectivity issues, to ensure that it continues to operate properly. Overall, implementing a home automation system based on PLC requires careful planning and execution to ensure that the system is functional, reliable, secure, and user-friendly.

4 Simulation and hardware connection

4.1 PLC programming and implementation process

Coding and the working principle are both critical aspects of a PLC-based project. The programming of a PLC involves writing code that determines how the PLC will interact with the machines or processes it is controlling. The working principle, on the other hand, refers to the underlying principles and mechanisms that allow the PLC to perform its intended functions. The coding determines how accurately and reliably the PLC will execute its tasks. The coding should be precise and error-free to ensure that the system performs as intended. The programming of a PLC can be customized to meet specific requirements of a project. The coding can be modified to optimize the performance of the system, improve efficiency, and reduce downtime. The working principle of a PLC-based system is designed to be scalable. As the system grows, it can be expanded with additional PLCs or input/output modules. In conclusion, coding and the working principle are critical components of any PLC-based project. Proper programming ensures the accuracy, reliability, and safety of the system, while the working principle allows for customization, maintenance, and scalability.

4.1.1 Enhanced energy efficiency part

In home automation systems, PLCs are used to control and monitor various devices and appliances such as lighting and home appliances. One important aspect of home automation is energy efficiency, and PLCs can play a crucial role in achieving this goal. By integrating sensors and other devices with the PLC, it can monitor the energy usage of various appliances and optimize their operation to reduce energy consumption. For example, a PLC can turn off lights when a room is unoccupied or adjust the temperature of a room based on occupancy and ambient temperature. Additionally, PLCs can provide real-time energy usage data to the homeowner, allowing them to make informed decisions about their energy usage and identify areas where they can further improve energy efficiency. With the ability to automate and optimize the operation of various devices and appliances in the home, PLCs can significantly enhance energy efficiency in home automation systems.

Window curtain part

sensors and relays to control the opening and closing of a window curtain based on the detection of sunlight. The system uses a photo sensor (I1) to detect the sunlight and two limit switches (I2 and I3) to detect if the window curtain is properly closed or open. The system also uses two relays (Q1 and Q15) to operate the motor that controls the opening and closing of the window

curtain.

When the photo sensor detects sunlight, the Q1 relay is turned on, which starts the motor in the forward direction to open the window curtain. Once the window curtain is fully open, the I2 switch detects it and turns off the Q1 relay, stopping the motor. When the sunlight is no longer detected by the photo sensor, the Q15 relay is turned on, which starts the motor in the reverse direction to close the window curtain. Once the window curtain is fully closed, the I24 switch detects it and turns off the Q15 relay, stopping the motor. In addition to the automated operation, the system also allows manual control of the window curtain using two push switches (I22 and I23) to open and close the window curtain manually.

As for the implementation process, it would involve connecting the sensors, relays, and motor to a microcontroller or PLC and programming the logic to control the opening and closing of the window curtain based on the input from the sensors and manual switches.

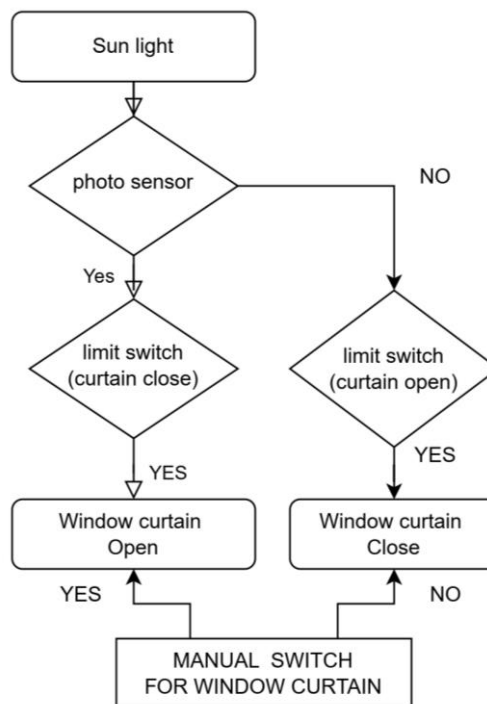


Figure 4.1 Flow chart for window curtain

Some challenges that could arise during the implementation process could include ensuring proper wiring and connections, calibrating the sensors to detect the appropriate light levels, and programming the logic to accurately control the motor and relays based on the input from the Sensors and switches.

Room light control

Room light control is an important feature of home automation as it provides a number of

benefits to the occupants of the house. Room light control can help reduce energy consumption by automatically turning off lights when they are not needed. With the use of occupancy sensors and timers, lights can be programmed to turn on or off based on the presence or absence of people in the room, saving energy and reducing electricity bills. With room light control, occupants of the house do not have to worry about manually turning lights on or off. The lights can be controlled remotely via a smartphone or voice command, providing a convenient and hassle-free experience. Room light control can be programmed to mimic natural daylight patterns, which can help regulate the body's circadian rhythm and improve sleep quality. By adjusting the color temperature and brightness of the lights, room light control can also help reduce eyestrain and headaches. In summary, room light control is an important feature of home automation that provides energy efficiency, convenience, security, personalization, and health benefits to the occupants of the house.

The implementation of the automated lighting system using the photo sensor I1 and Q10 relay involves several steps. First, the photo sensor I1 needs to be connected to the input of the PLC system. The Q10 relay, which will control the room light, needs to be connected to one of the output ports of the PLC.

Next, the PLC needs to be programmed to control the Q10 relay based on the input from the photo sensor I1. The program logic should include conditions to turn on the room light when the photo sensor I1 detects low light levels (i.e., when there is no sunlight and the window curtain is closed) and to turn off the room light when the photo sensor detects high light levels (i.e., when there is sunlight and the window curtain is open). Additionally, the logic should include a manual override option using the I3 SPDT switch to turn the room light on or off regardless of the photo sensor input.

One potential challenge in implementing this system could be ensuring that the photo sensor is positioned in a location that accurately reflects the light levels in the room. This may require careful placement and calibration of the sensor to ensure reliable operation. Additionally, the programming logic must be carefully designed to avoid false triggering of the light based on changes in lighting conditions that are not related to sunlight, such as turning on indoor lights or shadows passing over the photo sensor.

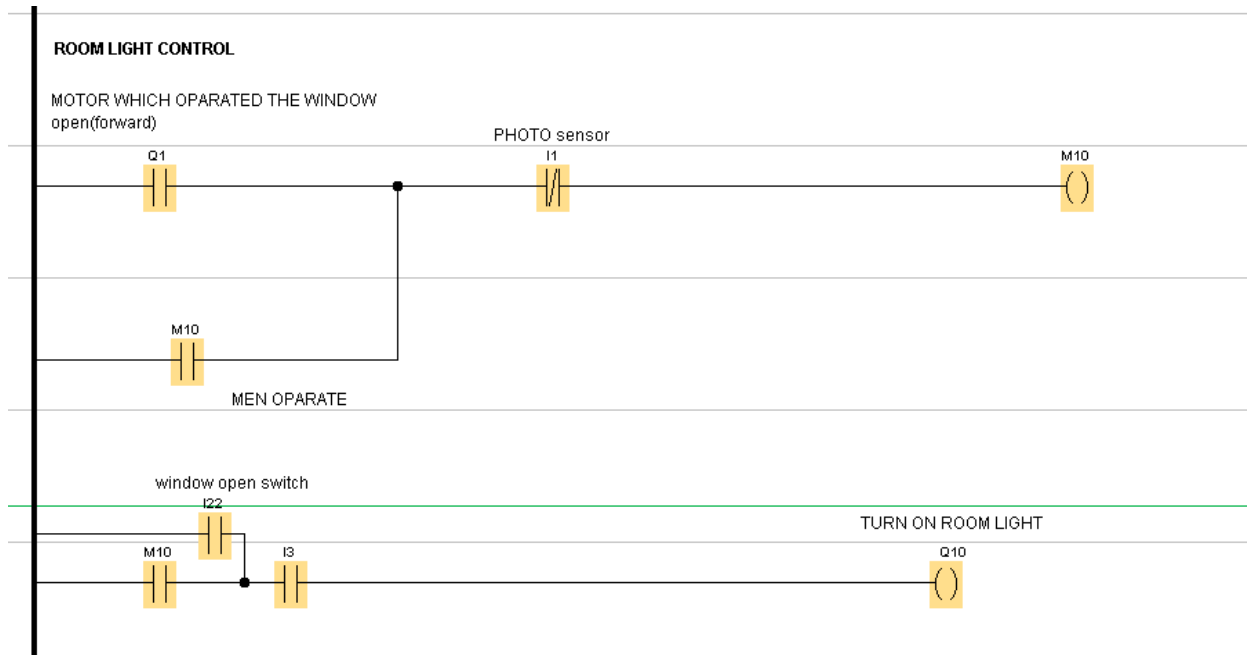


Figure 4.2 Ladder diagram for room light control

Main water pump control part

The main water pump control is an essential part of home automation as it helps to ensure the efficient and effective use of water in a household. The control system can be programmed to automatically turn on and off the main water pump based on the water demand and supply in the household. This helps to avoid unnecessary wastage of water, reduce energy consumption, and lower water bills. Moreover, the main water pump control can be integrated with sensors and other home automation systems to improve its functionality. For instance, it can be programmed to automatically shut off the water supply if there is a leak detected, reducing the risk of water damage and water wastage. Additionally, the main water pump control can be integrated with the irrigation system to automate watering the lawn or garden, ensuring that the plants receive adequate water at the right time. The main water pump control is an important part of home automation as it helps to ensure efficient use of water, reduce water bills, and prevent water damage. It also provides increased convenience and safety for homeowners.

Using a float switch to control the water level and turn on/off the main water pump is a common technique in automation. The float switch is a type of sensor that consists of a float and a switch. When the float moves up or down with the water level, it triggers the switch to turn on or off the water pump.

The output relay Q4 is used to control the main water pump. When the float switch detects a low water level, it will activate Q4 to turn on the water pump and fill the tank. When the water level reaches the high limit, the float switch will deactivate Q4 to turn off the water pump. This

automation helps to ensure that the water level is maintained within a certain range, preventing overfilling or running dry. It can also save energy by only running the pump when needed, rather than continuously.

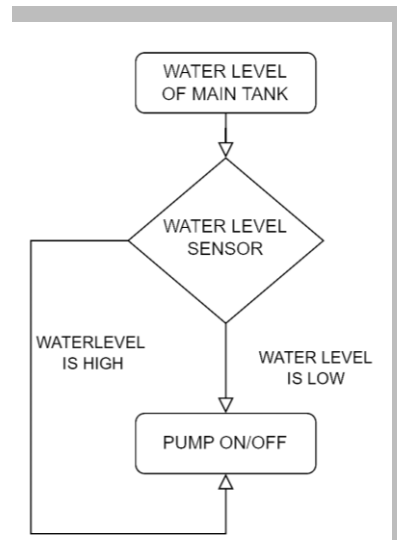


Figure 4.3 Flow chart for main water pump

Washroom light and ventilation control

In home automation, washroom light and ventilation control can be achieved using PLCs (Programmable Logic Controllers) and sensors. The system can be programmed to automatically turn on the lights and ventilation fan when someone enters the washroom and turn them off when the washroom is unoccupied. This can help save energy and reduce electricity bills. The washroom can be equipped with motion sensors that detect movement, enabling the PLC to turn on the lights and ventilation fan when someone enters the washroom. The ventilation fan can also be connected to a humidity sensor that activates the fan when the humidity level in the washroom exceeds a predetermined threshold. This can help prevent the buildup of moisture and the growth of mold and mildew, which can cause health problems. The washroom light and ventilation control system can also be integrated with the overall home automation system. For example, the system can be programmed to turn off the lights and ventilation fan when the house is empty or when the occupants are sleeping. This can help save energy and reduce the environmental impact of the home. Overall, washroom light and ventilation control in home automation can improve the energy efficiency and convenience of the home, while also providing health benefits by reducing the buildup of moisture and preventing the growth of mold and mildew.

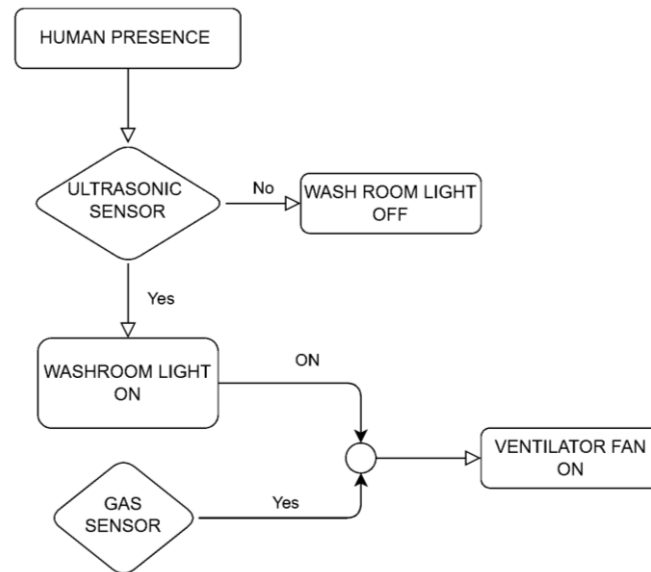


Figure 4.4 Flow chart for wash room ventilation

The implementation of the ultrasonic sensor I10 to control the lights and ventilation fan in the washroom is a common technique used in home automation systems. The ultrasonic sensor is mounted on the ceiling of the washroom and it detects the presence of a person based on the reflected sound waves.

When the sensor detects a person entering the washroom, it sends a signal to the microcontroller which turns on the Q7 relay to turn on the lights and the Q8 relay to turn on the ventilation fan. The lights can be connected to the Q7 relay and the ventilation fan to the Q8 relay.

Similarly, when the sensor detects that the washroom is unoccupied, it sends a signal to the microcontroller which turns off the Q7 relay to turn off the lights and the Q8 relay to turn off the ventilation fan. This saves energy by ensuring that the lights and ventilation fan are not left on unnecessarily when the washroom is not in use.

However, there can be challenges in implementing this system. One potential challenge is the placement of the ultrasonic sensor, as it needs to be mounted in a location that can detect the presence of a person reliably. Another challenge could be false triggers, where the sensor detects an object or movement other than a person and turns on the lights and fan. This can be addressed by adjusting the sensitivity of the sensor or using additional sensors to confirm the presence of a person.

4.1.2 Safety system part

Safety systems are an important aspect of home automation that can be implemented using a PLC. The safety system can be designed to detect potential hazards such as gas leaks, smoke, or fire, and take corrective action to prevent damage to property or harm to individuals. The PLC

can be programmed to monitor sensors and activate alarms or shut off appliances such as stoves or ovens when dangerous conditions are detected. The safety system can also be designed to provide remote monitoring and control, enabling homeowners to check the status of their home's safety system from a remote location. With a properly designed safety system, homeowners can have peace of mind knowing that their home is protected from potential hazards, even when they are away.

Fire detection system

A fire detection system is a crucial part of any home automation system, as it can quickly detect and alert occupants to the presence of a fire, allowing them to take immediate action to evacuate the building and call for emergency services. The fire detection system can be integrated into the home automation system, allowing it to be controlled and monitored remotely.

In the event of a fire, the fire detection system can trigger an alarm, alerting occupants to the emergency. The system can also be programmed to activate other components of the home automation system, such as turning on the lights or unlocking doors, to help occupants evacuate the building safely. Additionally, the fire detection system can be connected to a monitoring service or directly to the fire department, enabling emergency responders to be dispatched quickly in the event of a fire. The home automation system can also provide real-time updates to the homeowner or property manager, keeping them informed about the status of the fire detection system and any alerts or alarms that have been triggered. In summary, integrating a fire detection system into a home automation system can provide peace of mind for homeowners or property managers, by ensuring that they are quickly alerted to any potential fire hazards, and allowing them to take immediate action to protect themselves and their property. By automating the fire detection and alert process, the home automation system can provide an additional layer of safety and security to the building, ensuring the safety of the occupants and their belongings.

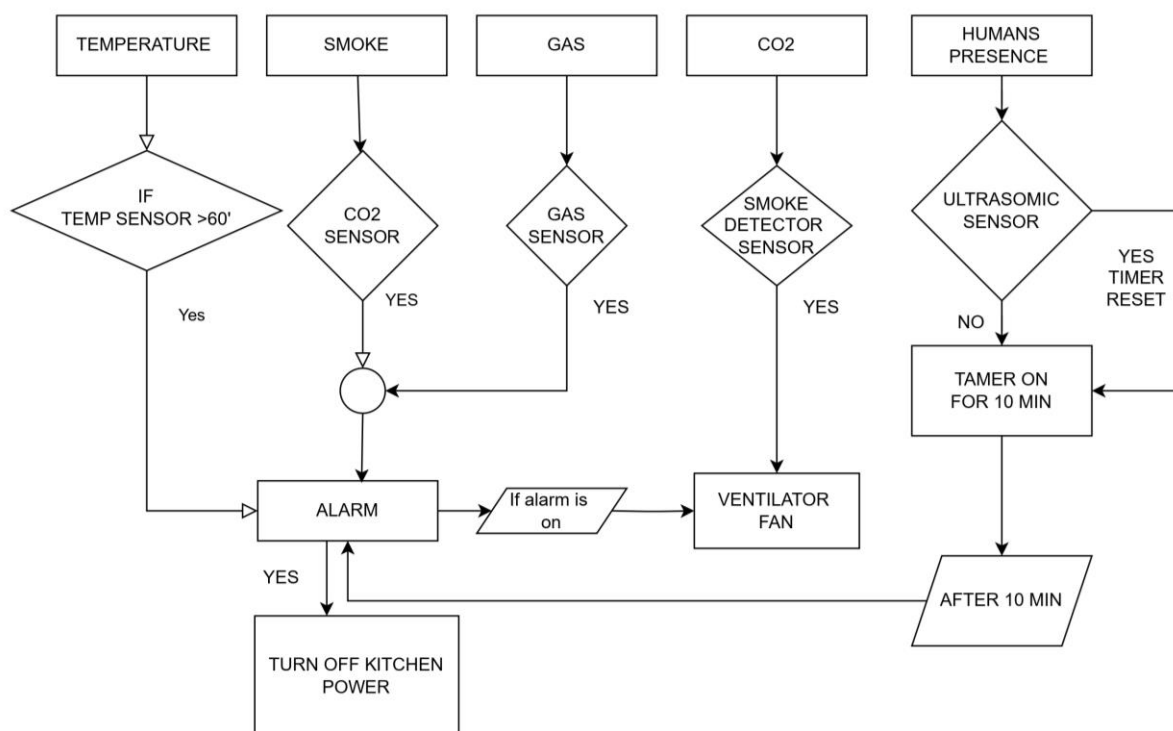


Figure 4.5 Flow chart for fire detection system

In this system, numerous sensors are utilized to monitor the living room and kitchen environment, with a special focus on kitchen safety. Due to the use of heavy electrical appliances that consume a lot of power and generate significant heat in the kitchen, an AI1 temperature sensor is employed to detect excess heat that could potentially cause a fire. If a fire does occur, the system can detect the increased levels of smoke and CO2 in the air through I9 smoke and I12 CO2 sensors, respectively. Additionally, a gas sensor is employed to detect gas leakage.

In the course of this research, it was discovered that people often forget to turn off kitchen appliances that generate significant amounts of heat after cooking. To address this issue, an ultrasonic sensor is utilized as I8 input to detect the presence of humans in the kitchen. If the ultrasonic sensor detects that no one is present in the kitchen, the system will automatically cut off the power supply to the appliances to prevent potential fire hazards. An alarm circuit can also be activated using Q2 output relay in case of emergencies, alerting individuals to take necessary precautions.

To prevent a fire from spreading, a safety system has been implemented. In the event of a fire, the system will automatically activate the Q5 room ventilation to decrease the levels of smoke and CO2 in the room, and the Q3 water spray pump will be activated to extinguish the fire.

4.1.3 Improve security system:

When it comes to home automation systems, security is a top priority. Improving the security system part in a PLC-based home automation system involves implementing measures to prevent unauthorized access, monitoring the system for any suspicious activity, and ensuring that critical information is encrypted and secure. One way to achieve this is by implementing user authentication protocols such as biometric identification or password protection to prevent unauthorized access. The system should also be designed to detect any attempts at tampering, and to alert the homeowner or security personnel in case of a breach. Furthermore, the communication between the home automation system and any external devices, such as mobile phones or cloud servers, should be encrypted to prevent interception of sensitive data. Additionally, regular security audits and updates should be performed to ensure that the system is up-to-date with the latest security measures and best practices. By implementing these measures, a home automation system can provide reliable and robust security, ensuring the safety and privacy of the homeowner and their property.

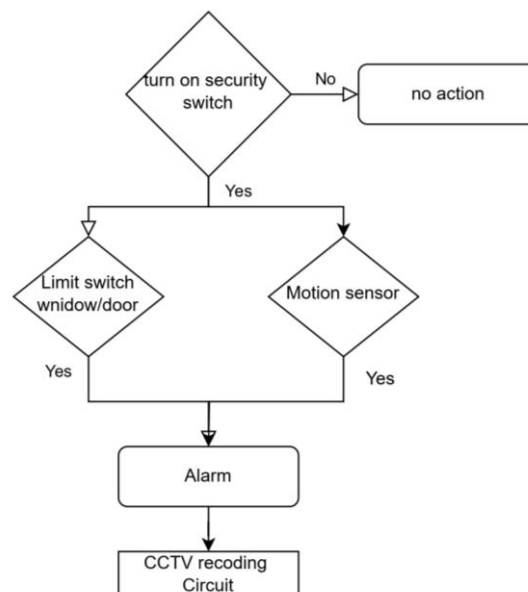


Figure 4.6 Flow chart for security system

The system mentioned above is equipped with a security feature that can be activated by a switch. Once the user activates this mode, the system becomes capable of detecting any unusual activity that may occur. The design of the system includes I13 and I14 limit switches that are used to detect whether windows or doors are open or closed. Additionally, a motion detector sensor, I15, is used to identify any unusual movement inside the room. If anyone tries to open a window or door while the security mode is on and the motion sensor detects any movement, it will activate the alarm circuit using the Q2 relay. Moreover, the CCTV circuit will start

recording automatically. This security system adds an extra layer of protection to the property, ensuring that it is secure and safe from any potential security breaches.

4.1.4 Improving comfort

Home automation using PLCs (Programmable Logic Controllers) can significantly improve the comfort of a home. The system can be programmed to automatically adjust the temperature, lighting, and other environmental conditions based on the occupants' preferences and activities. For example, the PLC can control the heating, ventilation, and air conditioning (HVAC) system to maintain a comfortable temperature in each room, based on the time of day and the occupancy of the room. Additionally, the PLC can be programmed to control the lighting system, adjusting the brightness and color temperature of the lights based on the time of day or the activities taking place in the room. The home automation system can also be integrated with sensors to detect motion, occupancy, or the opening and closing of doors and windows, enabling the PLC to adjust the environmental conditions accordingly. Overall, home automation using PLCs can enhance the comfort of a home by automating and optimizing the control of the environmental conditions, providing a personalized and convenient experience for the occupants.

HVAC system

HVAC stands for Heating, Ventilation, and Air Conditioning. It refers to the technology and systems used to provide heating, cooling, and ventilation to buildings, including homes, offices, and industrial spaces. HVAC systems are designed to maintain comfortable and healthy indoor air quality while also improving energy efficiency. HVAC systems can provide both heating and cooling, allowing occupants to maintain a comfortable temperature inside the building, regardless of the weather conditions outside. HVAC systems can filter and circulate indoor air, removing pollutants, allergens, and other contaminants, which can improve indoor air quality and help reduce health risks. Modern HVAC systems are designed to be energy-efficient, using less energy to operate, which can help reduce energy bills and minimize the carbon footprint of the building. By maintaining a consistent temperature and improving indoor air quality, HVAC systems can improve the comfort of the occupants of the building, creating a more productive and healthier environment. Regular maintenance of HVAC systems can help prolong the life of the equipment, reducing the need for costly repairs or replacements. Overall, HVAC systems are essential components of modern buildings, providing comfort, health benefits, and energy efficiency.

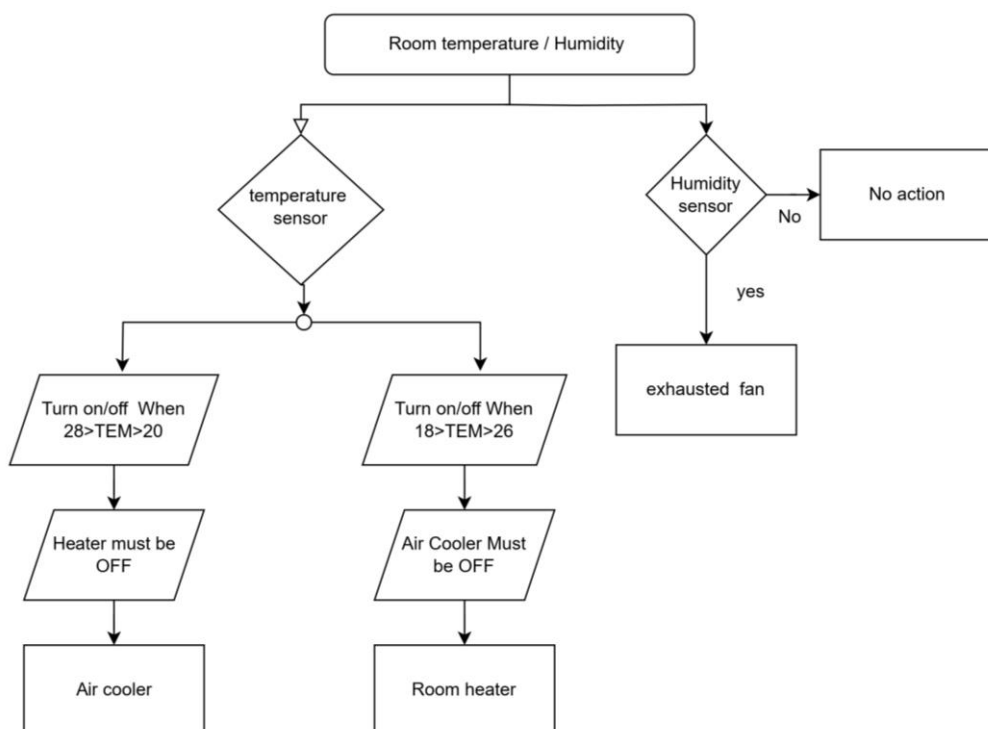


Figure 4.7 Flow chart for HVAC system

In the HVAC system, the A12 temperature sensor is used to measure the room temperature. Based on the temperature reading, the system can automatically turn on and off the air cooler or room heater by using Q11 or Q12 relay, respectively, to maintain a comfortable room temperature. Additionally, the system includes a humidity sensor to measure the humidity level of the room. If the humidity level is too low, the ventilation system will be turned on automatically to increase the humidity level to a comfortable level. This system provides users with a comfortable and healthy living environment by maintaining optimal temperature and humidity levels in the room.

Automatic pump control in garden

Home automation using PLCs (Programmable Logic Controllers) can be used to automate and control the water pump in a garden. With the use of sensors and timers, the PLC can control the water supply to the garden, ensuring that the plants are watered on a regular basis while minimizing water waste. The PLC can also monitor the water level in the storage tank and activate the pump as needed to maintain an adequate water supply. The system can be set up to automatically water the garden at specific times of the day or based on the moisture level of the soil, which can be measured using sensors. The PLC can be programmed to adjust the watering schedule based on weather conditions, such as rainfall, to avoid over-watering the plants. Furthermore, the PLC can be integrated with other home automation systems to improve the efficiency and convenience of the pump control system. For instance, the PLC can be connected

to a weather station to receive real-time data on rainfall and temperature, allowing it to adjust the watering schedule accordingly. The system can also be controlled remotely, using a smartphone or computer, allowing the user to monitor and adjust the watering schedule from anywhere. Overall, home automation using PLCs can provide an efficient and convenient way to control the water pump in a garden, improving the health of the plants while minimizing water waste and reducing the time and effort required for manual watering.

Using a humidity sensor as I20, the system can detect the humidity level of the soil. If the sensor detects that the humidity level is low, it can automatically turn on the water pump for several minutes to increase the humidity level. This can help ensure that the plants in the soil receive the appropriate amount of moisture, which is essential for their growth and overall health. By automating this process, the system can provide a consistent level of care for the plants, even when the user is not available to manually water them.

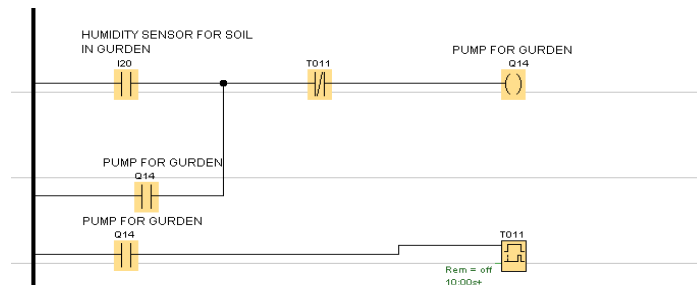


Figure 4.8 Ladder diagram for Garden

4.2 Hardware connection design

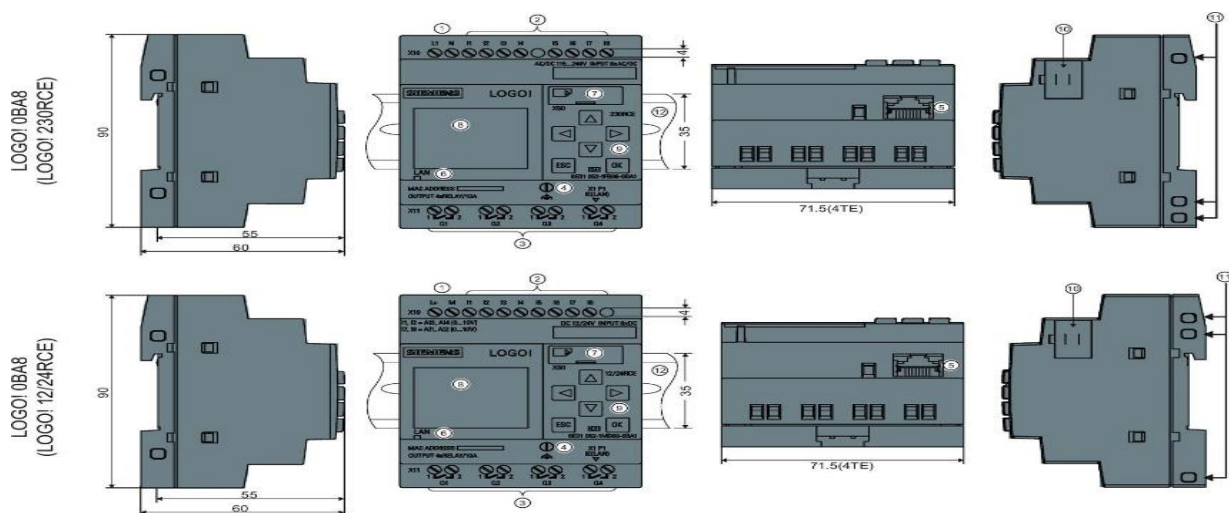


Figure 4.9 LOGO! Structure

1. Power supply
2. Inputs
3. Outputs

4. FE terminal, for connecting to earth ground
5. RJ45 interface, for connection to Ethernet (10/100 Mbit/s)
6. Ethernet communication status LED
7. Micro SD card slot
8. LCD
9. Control panel
10. Expansion interface
11. Mechanical coding sockets
12. Standard DIN rail

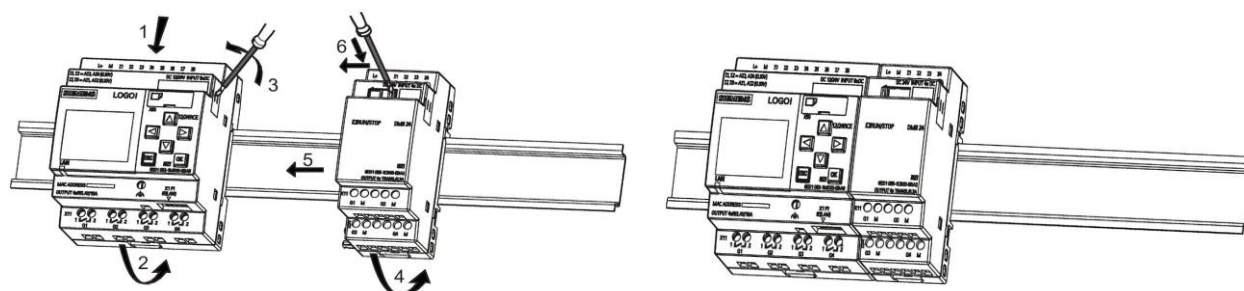


Figure 4.10 PLC mounting

To mount a LOGO! Base Module and a digital module onto a DIN rail, follow these steps:

1. Hook the LOGO! Base Module onto the rail.
2. Push down the lower end to snap it on. The mounting interlock at the rear must engage.
3. On the right side of the LOGO! Base Module/LOGO! expansion module, remove the connector cap.
4. Place the digital module onto the DIN rail on the right-hand side of the LOGO! Base Module.
5. Slide the digital module towards the left until it contacts the LOGO! Base Module.
6. Using a screwdriver, push the interlock to the left. In its end position the slide interlock engages in the LOGO! Base Module.

4.3 Connecting the power supply

The 230 V versions of LOGO! are suitable for operation with rated voltages of 115 VAC/VDC and 240 VAC/VDC. The LOGO! 24 V and 12 V versions can be operated with a 24 VDC, 24 VAC or 12 VDC power supply. For information on permissible voltage tolerances, line frequencies and power consumption, refer to the installation instructions in the Product Information supplied with your device and to the technical data in Appendix A.

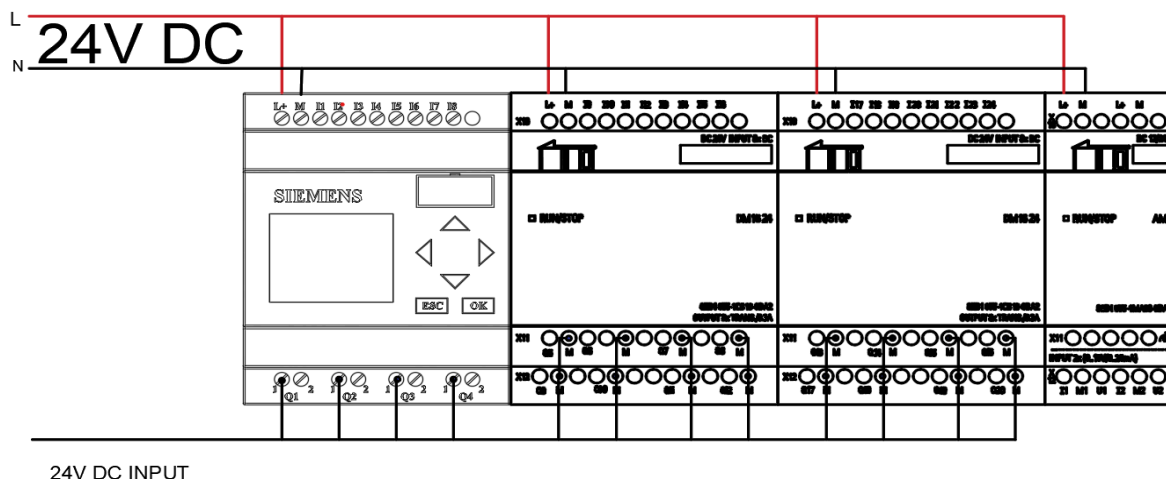


Figure 4.11 power connection

The LOGO! TDE must be supplied with a voltage of 12 VDC or 24 VAC/VDC.

4.4 Main connection diagram

The design of connecting sensors and equipment to a PLC-based intelligent home control system is crucial for seamless automation. It involves sensor selection, establishing wiring connections, choosing communication protocols, programming the PLC logic, integrating user interfaces, and thorough testing. This design ensures efficient data exchange and control between sensors, actuators, and the PLC, enhancing the automation, comfort, and convenience of the smart home.

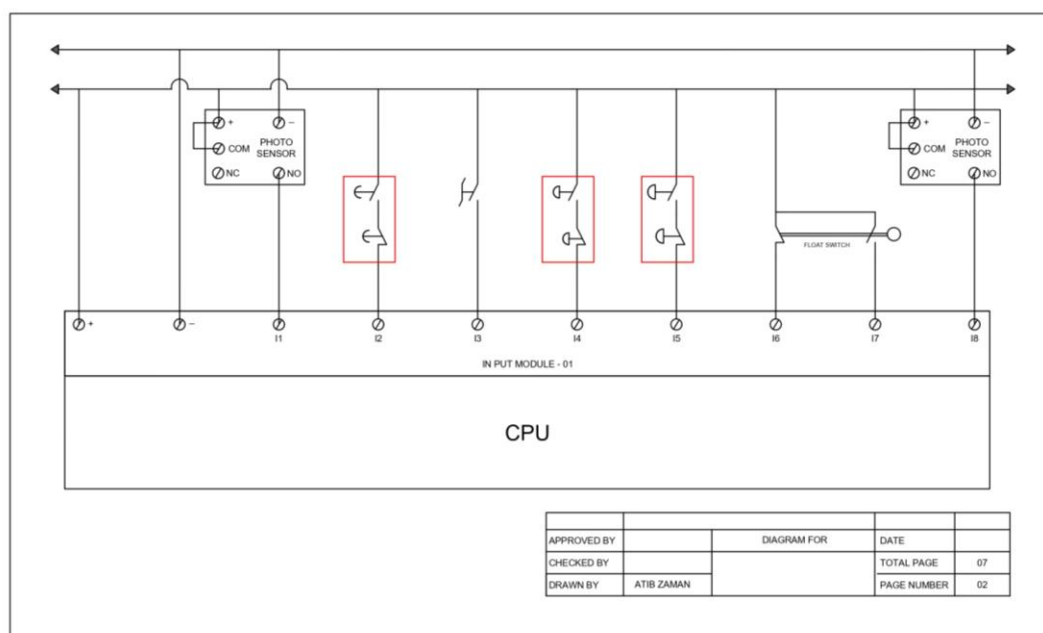


Figure 4.12 Input signal connection I1 to I8.

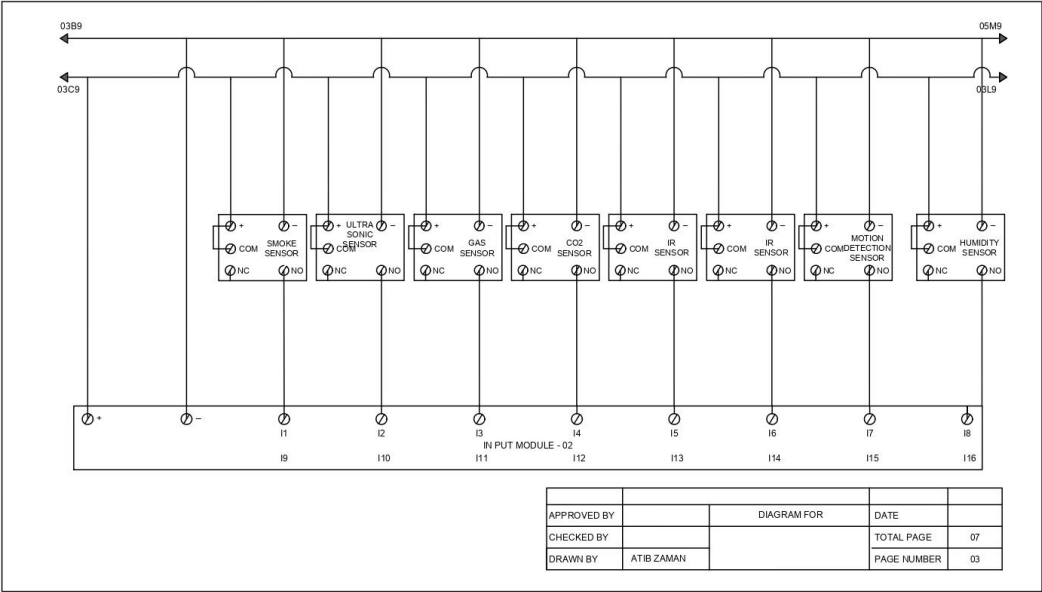


Figure 4.13 Input signal connection I9 to I16.

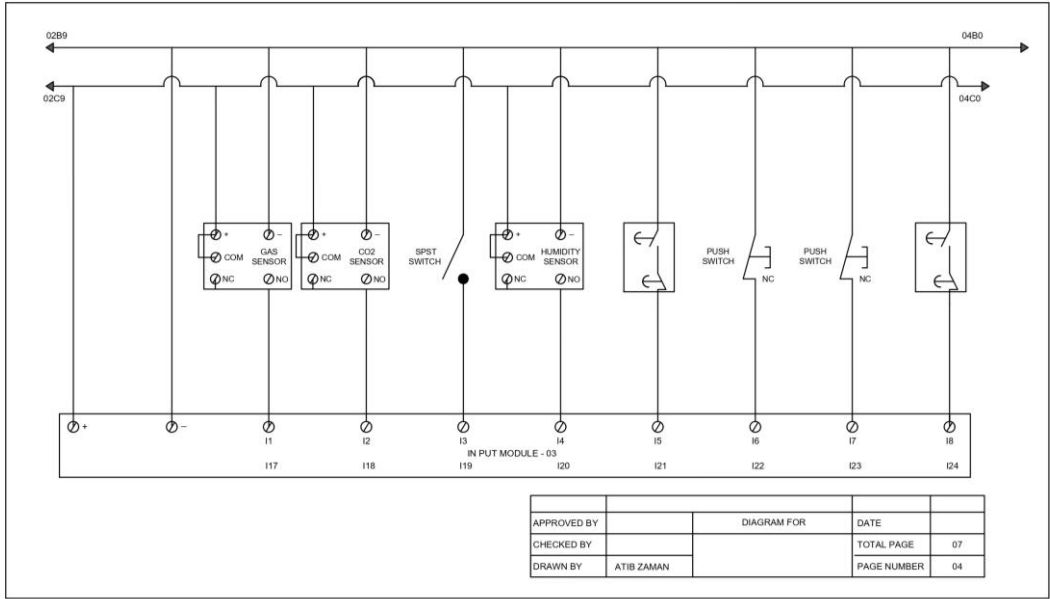


Figure 4.14 Input signal connection I17 to I24.

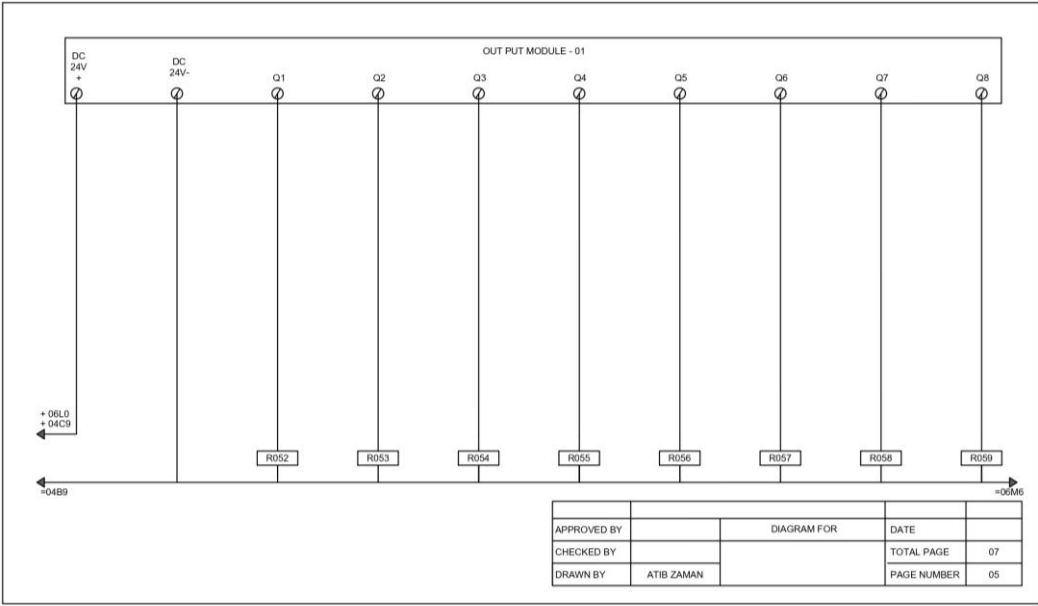


Figure 4.15 Output relay connection Q1 to Q8

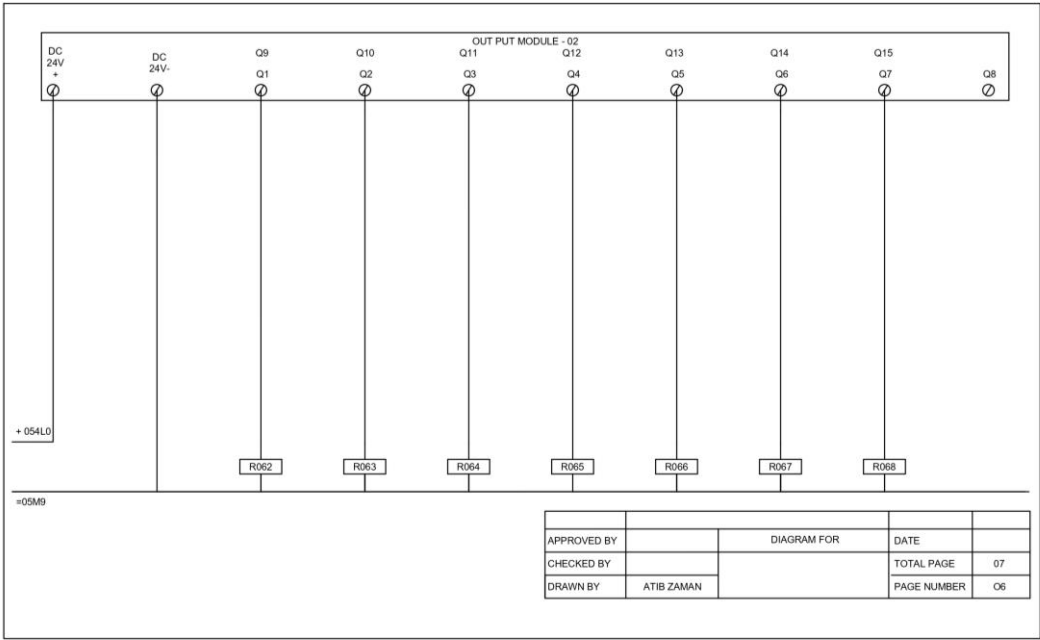


Figure 4.16 Output relay connection Q9 to Q15

5 Discussion

Home automation systems that are commonly used to control various household appliances, such as lighting, curtains, ventilation fans, and kitchen appliances. The systems employ different types of sensors, such as photo sensors, ultrasonic sensors, and temperature sensors, to detect environmental conditions and human presence. They also use relays to control the opening and closing of window curtains, turning on and off of lights and ventilation fans, filling and draining of water tanks, and cutting off power supply to kitchen appliances. The automation systems are designed to save energy, increase efficiency, and enhance safety and security by preventing potential fire hazards, reducing energy waste, and detecting any unusual activity through motion detectors and CCTV cameras. Logo PLCs are programmable logic controllers developed by Siemens, which are commonly used in small-scale automation applications such as home automation. These controllers are capable of controlling various devices, such as lights, motors, and sensors, based on pre-defined logic and input signals.

One potential implication of using a Logo PLC in home automation is the ability to create custom automation sequences tailored to the specific needs and preferences of the homeowner. For example, the PLC could be programmed to turn on specific lights when motion is detected in a particular room, or to adjust the temperature based on the time of day or weather conditions. Another potential benefit of using a Logo PLC in home automation is the ability to monitor and track energy consumption. By integrating sensors and meters into the automation system, the PLC can collect data on energy usage and provide insights into where energy is being wasted or could be conserved.

Additionally, using a Logo PLC in home automation can provide greater flexibility and scalability compared to traditional wiring and circuitry. With a programmable controller, it is easier to add or modify automation sequences as needed, without requiring extensive rewiring or infrastructure changes. Overall, the use of a Logo PLC in home automation can provide greater convenience, efficiency, and control for homeowners, while also offering potential cost savings through energy monitoring and conservation.

Home automation is a field that involves the integration of various technologies and systems to automate and control various aspects of a home, such as lighting, temperature, security, and entertainment, among others. PLC-based systems, on the other hand, are control systems that use programmable logic controllers (PLCs) to automate industrial processes and other applications.

In the context of home automation, PLC-based systems can be used to control and automate various aspects of a home, including lighting, heating, ventilation, air conditioning (HVAC), and security systems, among others. By using PLC-based systems, homeowners can enjoy greater

control over their home environment, leading to increased energy efficiency, comfort, and convenience.

The contribution of PLC-based systems to the field of home automation lies in their ability to provide reliable and flexible control over various home systems. PLCs are programmable and can be easily reconfigured to suit changing home automation needs, making them ideal for home automation applications. Additionally, PLC-based systems offer greater control and automation capabilities than traditional hard-wired systems, which require extensive rewiring to make changes. PLC-based systems have made a significant contribution to the field of home automation by providing a flexible and reliable means of controlling and automating various aspects of a home environment. As the demand for home automation continues to grow, PLC-based systems are likely to play an increasingly important role in the design and implementation of modern home automation systems.

5.1 Limitations and future research Area

Integration with emerging technologies: As new technologies such as Internet of Things (IoT) and Artificial Intelligence (AI) become more prevalent, there is a need to integrate them with PLC-based systems to enhance automation and improve the user experience. **Interoperability:** There is a need to ensure that different systems can communicate and work together seamlessly. This will require the development of standards for home automation systems to enable interoperability between different devices and systems.

Security: As home automation systems become more prevalent, there is a need to ensure that they are secure from cyber threats. Researchers need to investigate methods for securing these systems and protecting them from hacking.

User interface design: The user interface plays a crucial role in the usability and adoption of home automation systems. There is a need for research on designing user interfaces that are intuitive, easy to use, and accessible to people with disabilities.

Energy efficiency: Home automation systems can be designed to optimize energy consumption and reduce waste. There is a need for research on developing energy-efficient home automation systems that can help reduce energy costs and promote sustainability.

Cost: The cost of home automation systems can be a barrier to adoption, particularly for low-income households. There is a need for research on developing affordable home automation systems that can be accessible to a wider range of households. The field of home automation and PLC-based systems is still evolving, and there is a need for continued research and development to address these and other challenges.

6 Conclusion

The present study focused on the development and implementation of a smart home automation system based on Logosoft PLC. Under the guidance of my course supervisor, Ling Lu, at China Three Gorges University, this research aimed to investigate the potential of integrating Logosoft PLC into residential environments to create an efficient and convenient smart home experience.

The study commenced with an in-depth literature review, exploring the existing literature and research on smart home automation systems, PLC technology, and their applications. This phase provided a comprehensive understanding of the theoretical background and enabled the identification of research gaps in the field. To accomplish the objectives, a practical approach was adopted, including system design, hardware and software implementation, and performance evaluation. Logosoft PLC was selected as the core control unit due to its flexibility, reliability, and compatibility with various home automation devices.

The implemented smart home automation system demonstrated remarkable outcomes and contributions. Firstly, it successfully achieved the automation of various household appliances, such as lighting, heating, air conditioning, and security systems, leading to enhanced energy efficiency, cost savings, and improved comfort for residents. The system's user-friendly interface allowed easy control and monitoring of devices remotely via mobile applications, enabling convenient management of home environments. Furthermore, the study contributed to the body of knowledge in the field of smart home automation by presenting a detailed analysis of the system's performance. Through extensive experimentation and data collection, the system's response time, reliability, and energy consumption were evaluated, providing valuable insights into its operational efficiency.

Overall, the findings of this study validate the feasibility and effectiveness of integrating Logosoft PLC into a smart home automation system. The research contributes to the growing field of home automation by offering practical insights, addressing research gaps, and providing a foundation for further advancements in the area. It is anticipated that the outcomes of this study will benefit both researchers and practitioners in the field of automation, facilitating the development of more sophisticated and intelligent home environments in the future.

In conclusion, the study highlights the effectiveness of using a Logo PLC-based system for home automation. The results indicate that the system is reliable, efficient, and provides a high level of control for home appliances. This study's findings have significant implications for both the home automation industry and the wider field of PLC-based systems.

Acknowledgement

I would like to express my deepest gratitude to my thesis supervisor, Associate Professor Lu Ling, for her meticulous guidance and unwavering support throughout the entire process of my graduation thesis. Her rigorous academic approach and scientific methods have significantly influenced my work. Despite her busy schedule, she provided careful guidance at every stage, from topic selection to data review, thesis outline determination, mid-term thesis modification, and final formatting adjustments. Her invaluable advice and attention to my future direction have been instrumental in shaping this thesis. I extend my sincere thanks and deep respect to her. I am grateful to the friends and fellow students who provided enthusiastic support and assistance with the theoretical research work in my thesis. Their contributions have been invaluable. Furthermore, I want to express my appreciation to all the teachers who have imparted their knowledge and expertise over the past four years. They have played a significant role in shaping my academic journey and have contributed to the successful completion of this graduation thesis. I also want to thank my classmates for their support and assistance throughout these years, as they have taught me the importance of professional knowledge, learning, and innovation. I extend my sincere gratitude to them and wish them success in their future endeavors.

This graduation project has provided me with the opportunity to integrate theory with practice and develop comprehensive skills in electrical design and related theories. It has enhanced my ability to address real-world engineering problems and furthered my knowledge in electrical control. Writing this graduation thesis has been a systematic learning process, marking the beginning of a new chapter in my educational journey. I will always remember my time as a student at China Three Gorges University (CTGU) and will uphold the school motto of the "spirit of seeking" in my future endeavors.

Finally, I would like to extend my utmost respect and sincere thanks to all the teachers, friends, and classmates who have supported and helped me along the way. Your encouragement and assistance have been invaluable, and I am truly grateful for your contributions.

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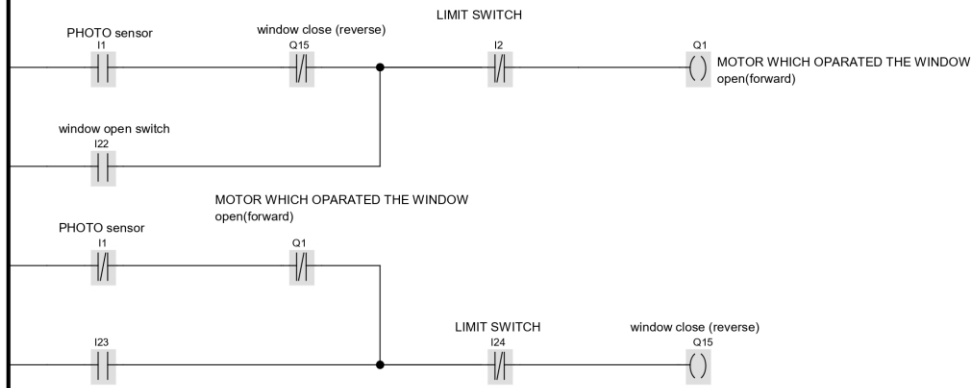
Appendix

NO	MEMORY ADDRESS	PORT TYPE	SWITCHING TYPE	WORKING PRINCIPLE
1	I1	INPUT	photo sensor	detect sunlight
2	I2	INPUT	limilimit switch	detect thhe window is close properly
3	I3	INPUT	SPST switch	manual switch for room light
4	I4	INPUT	push switch	for turn of the emergency alarm.
5	I5	INPUT	emergency switch	manual trun of switch for emergency system
6	I6	INPUT	float switch	turn on when the water level is low
7	I7	INPUT	float switch	turn of when the water level is high
8	I8	INPUT	untra sonic sensor	for detect human present inside the kitchen
9	I9	INPUT	smoke sensor	detect smoke present inside the kitchen
10	I10	INPUT	ultra sonic sensor	detect human presence in wash room
11	I11	INPUT	gas sensor	detect gas leakge inside the kitchen
12	I12	INPUT	CO2 detect sensor	detect co2 contains inside the kitchen
13	I13	INPUT	IR sensor	active if someone trying to open the window
14	I14	INPUT	IR Sensor	active if someone trying to open the door
15	I15	INPUT	motion detect sensor	motion detection inside the room
16	I16	INPUT	humidity sensor	indicate humidity lavel inside the living room
17	I17	INPUT	gas sensor	indicate gas level inside the living room
18	I18	INPUT	co2 sensor	indicate co2 lavel inside the living room
19	I19	INPUT	switch	turn on security system
20	I20	INPUT	humidity sensor	for detect the humidity lavel of garden soil
21	I21	INPUT	limit switch	limit switch for detect the window is close now
22	I22	INPUT	push switch	for opening the window
23	I23	INPUT	push switch	for closing the window
24	I24	INPUT	limit switch	detect the window is close now
25	Q1	OUTPUT	Relay	turn on window curtain oparated motor (forward)
26	Q2	OUTPUT	Relay	alarm
27	Q3	OUTPUT	Relay	water pump for protection
28	Q4	OUTPUT	Relay	water pump
29	Q5	OUTPUT	Relay	kitchen power supply
30	Q6	OUTPUT	Relay	ventilation fan for kitchen

31	Q7	OUTPUT	Relay	turn on wash room light
32	Q8	OUTPUT	Relay	turn on ventilation fan inside the living room
33	Q9	OUTPUT	Relay	start the CCTV recording circuit
34	Q10	OUTPUT	Relay	turn on room light
35	Q11	OUTPUT	Relay	turn on air cooler
36	Q12	OUTPUT	Relay	turn on room heater
37	Q13	OUTPUT	Relay	turn on room ventilation
38	Q14	OUTPUT	Relay	water pump for garden
39	Q15	OUTPUT	Relay	turn on window curtain operated motor reverse
40	AI1	analog input	Pt 100	for detect kitchen temperature
41	AI2	analog input	Pt 100	for detect room temperature

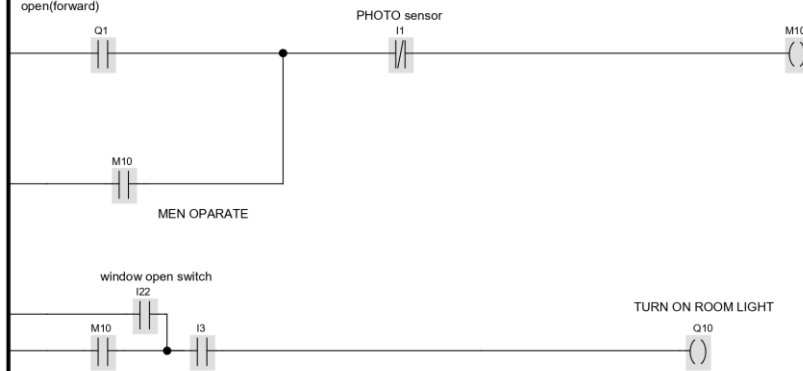
ENHANCED ENERGY EFFICIENCY PART

window control part

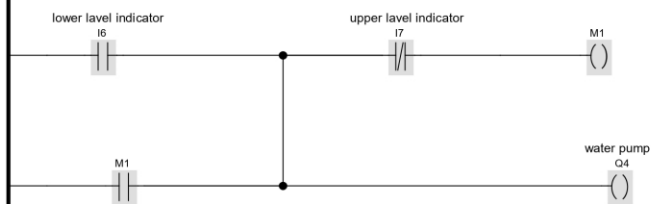


ROOM LIGHT CONTROL

MOTOR WHICH OPERATED THE WINDOW
open(forward)



water pump part



Creator:	jaman	Project:		Customer:	
Checked:		Installation:		Diagram No.:	
Date:	5/5/23 6:47 PM/6/11/23 1:07 AM	File:	Circuit Diagram2.Ild	Page:	1 / 4

